

Case Study 08: Design and Development of a Patient-Specific Zygomaticomaxillary Reconstruction Implant

Alternative names:

- Zygomatic Reconstruction Implant
- Midface Reconstruction Implant
- Zygomatic-Orbital Reconstruction Implant

1. Introduction

The zygomatic bone (cheek bone) and infraorbital region are important structures responsible for facial contour, cheek projection, orbital support, and facial symmetry. Large defects in this region may occur due to:

- Mucormycosis (fungal infection)
- Tumor resection
- Severe facial trauma
- Maxillofacial defects
- Congenital deformities

Conventional reconstruction plates often fail to restore the complex anatomy of the zygomatic region. Therefore, Patient-Specific Implants (PSIs) are designed using CT scan data to achieve accurate anatomical reconstruction and improved aesthetic outcomes.

2. Objective

The objective of this work was to design and develop a patient-specific implant for reconstruction of the zygomaticomaxillary region, restoring:

- Facial projection
- Facial width
- Orbital support
- Facial symmetry
- Aesthetic appearance

3. Implant Description

The designed implant is a customized titanium implant consisting of:

3.1 Superior Fixation Arm

Provides fixation to the frontal bone and lateral orbital rim.

3.2 Inferior Fixation Arm

Provides fixation to the zygomatic body and maxillary region.

3.3 Central Bulged Region

Restores the natural cheek projection and facial contour.

3.4 Screw Fixation Zones

Multiple screw holes provide stable fixation and load distribution.

The implant exhibits a C-shaped anatomical geometry that closely follows the patient's zygomatic anatomy.

4. Design Features

Patient-specific anatomical fit

Restoration of facial projection

Restoration of facial width

Reconstruction of zygomatic contour

Multiple fixation points

Smooth anatomical transition

Lightweight design

Suitable for additive manufacturing

5. Methodology

1. Acquisition of 1 mm slice CT DICOM data.
2. Segmentation and 3D skull reconstruction.
3. Identification of the defect region.
4. Mirror reconstruction of the healthy side.

- 5. Generation of patient-specific implant geometry.
- 6. Thickness generation.
- 7. Creation of screw holes.
- 8. Surface smoothing and refinement.
- 9. STL export for manufacturing.

6. Material Selection

Material: Titanium Alloy (Ti-6Al-4V ELI Grade 23)

Advantages

- Excellent biocompatibility
- High corrosion resistance
- High strength-to-weight ratio
- Good osseointegration
- Suitable for SLM manufacturing

7. Mechanical Properties

Property	Cortical Bone	Titanium
Modulus of Elasticity	8–24 GPa	103 GPa
Yield Strength	115 MPa	≥230 MPa

8. Typical Design Specifications

Parameter	Value
Implant Type	Patient-Specific Zygomaticomaxillary Reconstruction Implant
Material	Ti-6Al-4V ELI Grade 23
Manufacturing Method	SLM / LPBF
Thickness	1.0–1.5 mm

Screw Hole Diameter	2.0–2.3 mm
Screw Size	2.0 mm CMF screws
Surface Finish	Polished
File Format	STL

9. Clinical Applications

- Mucormycosis defects
- Zygomatic bone loss
- Tumor resection defects
- Severe facial trauma
- Maxillofacial reconstruction
- Secondary facial reconstruction surgeries

10. Advantages of Patient-Specific Implant

- Better anatomical fit
- Reduced operation time
- Improved facial aesthetics
- Reduced surgical complexity
- Better restoration of facial contour and symmetry

11. Conclusion

A patient-specific zygomaticomaxillary reconstruction implant was successfully designed to restore the anatomy of the zygomatic region. The implant closely replicates the original facial contour and provides stable fixation through multiple screw holes. The design is suitable for fabrication using Titanium Ti-6Al-4V ELI through Selective Laser Melting (SLM) technology and can effectively restore both function and aesthetics in patients with large zygomatic defects.

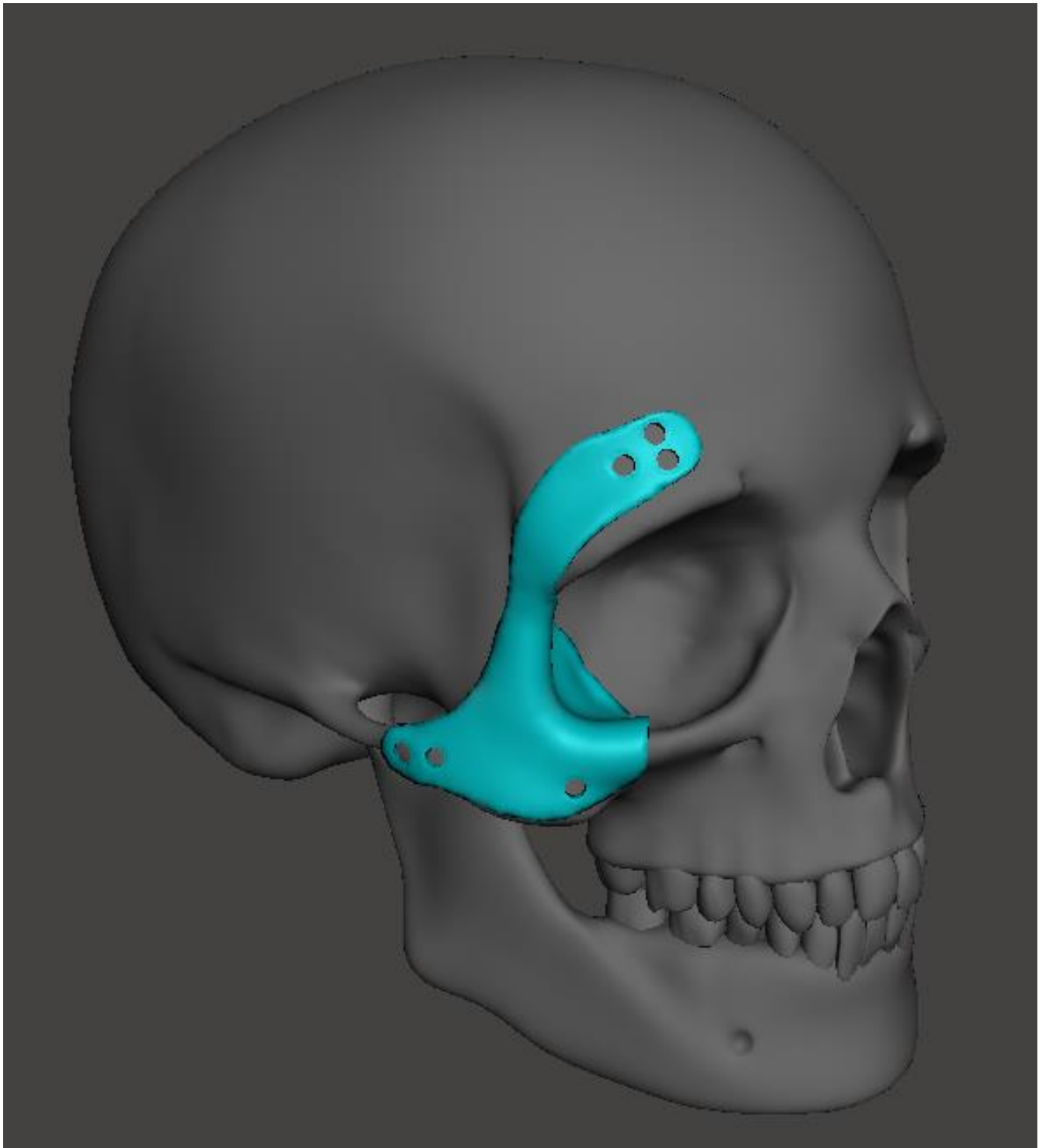


Figure X. Patient-Specific Zygomaticomaxillary Reconstruction Implant designed for reconstruction of a zygomatic defect and restoration of facial contour and projection.

